stage_release tool

Franklin D.

Abstract

This is a custom tool written for a specific team at a certain company. The tool can be used to "stage" software releases on a large number of PANOS devices with the goal of minimizing operator typing and machine interaction.

stage_release

Panorama XML API

In order to allow the BASH script to communicate with Panorama via the XML API, you need to enable the API access.

As best practice, set up a separate admin account for the XML API access.

1. Configure an Admin Role Profile:
   - Select `Device` > `Admin Roles` and click `Add`.
   - Enter a `Name` to identify the role.
   - Select the `Role` scope.
   - In the `XML API` tab, click the icon for each functional area to toggle it to the desired setting: `Enable` or `Disable`.
   - Click `OK` to save the profile.
   - Assign the role to an administrator - complete the `Configure a Firewall Administrator Account` step.
2. Configure a Firewall Administrator Account:
   - Select `Device` > `Administrators` and click `Add`.
   - Enter a user `Name` to identify the account. If the firewall uses a local user database to authenticate the account, enter the name that you specified for the account in the database.
   - Select an `Authentication Profile` or sequence if you configured either for the administrator. If the firewall uses `Local Authentication` without a local user database for the account, select `None` (default) and enter a `Password`.
   - Select the `Administrator Type` > `Custom Panorama Admin`.
   - Select the `Admin Role` you created for this account in the `Profile` section.

- Click `OK` and `Commit`.

Configure BASH for XML API

Export these environment variables to prepare for script execution.

NOTE: You may also specify USER and PAN_IP by passing flags to the script.

```
export USER=<the_name_of_the_firewall_administrator_account_you_configured> # OPTIONAL
export PASS=<the_password_of_the_firewall_administrator_account_you_configured>
export PAN_IP=<the_panorama_instance_ip_address> # OPTIONAL
```

The following list can be generated by calling the script with the `-h` flag.

```
franklin@ups:~/bin$ ./stage_release.sh -h

stage_release script.

Syntax: stage_release.sh [-f|-h|-i|-p|-r|-t|-u|-v]
options:
-f     Specify a file with FW IPv4 addresses, one per line.
-h     Print this Help.
-i     Install release
-p     Use PROD environment
-r     Specify a sofwtware release to download/install (ex: 10.2.0)
-t     Use TESTNET environment
-u     Specify a USER for the Panorama
-v     Enable verbose (debug) mode.
```

Validate XML API configuration

To check if the `XML API` has been enabled, run the following code snippet:

```
curl -k -X GET 'https://$PAN_IP//api/?type=keygen&user=$USER&password=$PASS'
```

If everything was configured correctly, a response status `success` should be returned together with an API key for the specified account.

Tool Env Configuration

There are some Bash envrironment variables that can be set in order for the tool to function properly. These are listed in the following subsections.

OPTIONAL BASH ENV VAR: USER

This is the username you created previously for the dedicated XML API user.

NOTE: You may also specify a user from the command line with the `-u` flag.

For the `USER` setting, the tool has a default value of `xml-api-user`. To override this value, use the following command before running the tool:

```
export USER="my_new_username" # set the value
echo $USER # validate it was set properly
```

BASH ENVIRONMENT VARIABLE: PASS

This is the password for the new XML API user that was created. It is important that we do not hardcode any active passwords into the script.

For the `PASS` setting, the tool has a default value of `default`. To override this value, use the following command before running the tool:

```
export PASS="my_secret password" # set the value
echo $PASS # validate it was set properly
set | grep $PASS # different way to validate
```

OPTIONAL BASH ENV VAR: PAN_IP

This is the IPv4 address of the active Panorama.

NOTE:

- You may specify TESTNET Panorama IP from the command line with the `-t` flag.
- You may specify PRODUCTION Panorama IP from the command line with the `-p` flag.

For the `PAN_IP` setting, the tool has a default value of `34.134.31.136`. To override this value, use the following command before running the tool:

```
export PAN_IP="10.2.3.4" # example, set the value
echo $PAN_IP # validate it was set properly
set | grep $PAN_IP # another way to validate
```

Advanced

Alternatively, you may choose to use a combination of the direnv and pass tools to make the management of these variables slightly easier.

Execution

Create a text file and add in IP addresses, one per line. This will allow you to specify exactly which firewalls to operate on.

Example:

```
3.233.53.199
44.216.25.244
34.206.152.182
52.55.185.160
```

To run the script with a list of serials in a file, type the command `./stage_release.sh -f my_ip_list.txt`

To run the script against all firewalls that are managed by this Panorama, type the command `stage_release.sh`

Dev Notes

- Use `pandoc` to generate documentation

```
sudo apt install pandoc
cd docs && pandoc stage_release.md -o stage_release.pdf
```